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RESEARCH ARTICLE

Network Intrusion Detection Based on Feature Image and Deformable Vision Transformer Classification

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ABSTRACT Network intrusion detection technology has always been an indispensable protection mechanism for industrial network security. The rise of new forms of network attacks has resulted in a heightened demand for these technologies. Nevertheless, the current models' effectiveness is subpar. We propose a new Deformable Vision Transformer (DE-VIT) method to address this issue. DE-VIT introduces a new deformable attention mechanism module, where the positions of key-value pairs in the attention mechanism are selected in a data-dependent manner, allowing it to focus on relevant areas, capture more informative features, and avoid excessive memory and computational costs. In addition to using deformable convolutions instead of regular convolutions in embedding layers to enhance the receptive field of patches, a sliding window mechanism is also employed to utilize edge information fully. In Parallel, we use a layered focal loss function to improve classification performance and address data imbalance issues. In summary, DE-VIT reduces computational complexity and achieves better results. We conduct experimental simulations on the public intrusion detection datasets, and the accuracy of the enhanced intrusion detection model surpasses that of the Deep Belief Network with Improved Kernel-Based Extreme Learning (DBN-KELM). It reaches 99.5% and 97.5% on the CIC IDS2017 and UNSW-NB15 datasets, exhibiting an increase of 8.5% and 9.1%, respectively.

INDEX TERMS Network intrusion detection, deformable vision transformer, deformable convolution, deformable attention mechanism, vision transformer.

I. INTRODUCTION

With the rapid development of the Internet, massive multimedia information is being disseminated, which poses a significant challenge to Internet security due to the attacks of hackers and information leaks. Therefore, preventing network intrusion has become the central issue of network security. To address this problem, the development of intrusion detection technology has become necessary. Our main objective is to create a network intrusion detection system that is both

secure and effective. To date, numerous Intrusion Detection Systems (IDS) have been designed utilizing various technologies, including traditional visual analysis [1] and machine learning [2].

Supervised algorithms like Naive Bayes (NB) [4] and Support Vector Machine (SVM) [3] are commonly employed in traditional machine learning for solving classification problems. However, they must repeatedly label a substantial amount of data, which is time-consuming and laborious. On the other hand, unsupervised algorithms like Principal Component Analysis (PCA) [6] and Gaussian Mixture Model (GMM) [5] can discover hidden structures and information

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in the data without data labeling. However, their prediction accuracy is low, and evaluating the model's performance is challenging.

Visual analysis can transform abstract information into understandable image data, making it easier for people to discover patterns in network security data and alarm information. However, personal subjective factors can lead to different interpretations of the same data.

Deep learning algorithms such as Convolutional Neural Network (CNN) [7], Recurrent Neural Network (RNN) [8], and Generative Adversarial Network (GAN) [9] have their limitations. RNN is helpful for processing sequence data and can capture dependencies between sequences. However, in some cases involving long-term dependencies, deep learning models may encounter issues with gradient explosion and vanishing during training. It is not parallel computing, which affects training speed. GAN can solve imbalanced data problems in intrusion detection by using a generator and a discriminator. When the network layer is deep enough in CNN, it can parallelly compute and capture remote features. However, deep convolution's [10] effect is not always ideal, as learning remote dependency relationships mainly depends on the length of the path through which forward and backward signals pass in the network [11].

Vision Transformer (ViT) [12] addresses the issues of modeling global dependencies in inputs and outputs through attention mechanisms while avoiding recurrent structures and enabling parallel computing capabilities. By calculating the correlation between any two positions regardless of distance, the attention mechanism allows for better and faster feature extraction. Additionally, the ViT's multi-head attention mechanism obtains richer feature information by forming multiple subspaces and extensively utilizing information from different representation subspaces at different locations.

Based on the advantages above, we propose a new intrusion detection model called Deformable Vision Transformer (DE-ViT) tailored to the task. To achieve this, we have enhanced the ViT model, which has several advantages. Specifically, we have addressed the limitation of ViT ignoring local structures when directly segmenting images into fixed-length tokens by suggesting using a sliding window mechanism to segment the data. This approach improves edge information preservation. Additionally, we have introduced a deformable attention mechanism to avoid excessive memory and computational costs resulting from using dense self-attention in ViT. Furthermore, features may be affected by irrelevant parts beyond the region of interest. Specifically, having too many keys for each query patch will result in high computational costs, slow convergence, and an increased risk of overfitting.

In contrast, the convolution module [13] uses deformable convolution to enlarge the receptive field of each patch without increasing the convolution kernel. The model uses a hierarchical focal loss function to assign weights to samples

with different categories [14]. This approach effectively addresses prediction errors and reduces the impact of outliers on the model, thereby enhancing the classification performance. The simulation experiments conducted on the CIC IDS2017 [32] and UNSW-NB15 [33] datasets show that our proposed intrusion detection method has significantly improved accuracy.

Overall, the contributions of this paper are as follows:

Firstly, we propose DE-ViT, which improves upon traditional neural networks like CNN by addressing their poor performance in global information perception [15]. This improvement enhances the generalization capabilities of the detection model.

Secondly, we use a small convolution kernel and stride instead of the original hard segmentation. This enables better correlation modeling between each patch and its adjacent patches, allowing for better modeling of local information such as edges and lines.

Thirdly, we introduce a deformable attention mechanism [16] for computing the correlation between two positions. So that the model can extract features more effectively, focus on relevant regions, capture more information features, and reduce computation. Moreover, deformable convolution [17] increases the receptive field of each patch without increasing the convolution kernel size. This allows each patch to contain more information, resulting in a more efficient feature extraction process.

Fourthly, we add sine and cosine positional encoding to assign position information to each feature. This helps the model understand their relationships since different positions represent different features.

Finally, we use the hierarchical focal loss function to solve the data imbalance problem and improve the classification performance by reducing the impact of outliers on the model.

The rest of this article follows: Part II introduces the relevant information to be consulted. Part III proposes an improved DE-ViT intrusion detection model. Part IV provides binary and multiclass experimental results and compares them with other methods. Part V is the summary of the full text.

II. RELATED WORK

Security issues have recently plagued Network-based Intrusion Detection Systems (NIDS). Our focus is on finding solutions to these problems. Traditional machine learning and deep learning, particularly deep learning, are the preferred solutions. Deep learning outperforms traditional machine learning algorithms in processing data in different fields, such as natural language, image, and speech. It has also proven to be highly effective in intrusion detection. However, machine learning has its limitations. For instance, Mishra et al. and colleagues [23] conducted a comprehensive study and survey of different machine-learning algorithms used in intrusion detection. They outlined the limitations of various technologies. For example, If the data contains misclassified

instances, Support Vector Machines (SVM) [18] may fail to find the separating hyperplane. The Naive Bayes classifier [19] typically assumes attribute independence, which may not hold in practical applications. Decision trees [20] can be prone to overfitting if the model becomes overly complex and fits too closely to the training data, resulting in poor generalization ability to new data. Random forests [21] may have high computational costs and perform poorly on imbalanced datasets. Moreover, the K-Nearest Neighbors (KNN) algorithm [22] is affected by the selection of the distance measure and the value of K [23].

Nowadays, the rapid growth and expansion of computer networks and related applications have led to significant achievements in various network security tasks through deep learning. For instance, Khan et al. [24] proposed a new framework for Network Intrusion Detection System (NIDS) based on deep convolutional neural networks. They utilized network spectrum images generated by short-time Fourier transform. The objective of this solution is to overcome the challenge of high False Alarm Rates (FAR) encountered when detecting new attacks, which has been a persistent issue for existing NIDS solutions. To address this challenge, Lopez-Martin et al. [25] introduced a novel method for network intrusion detection. Their approach utilizes an extended radial basis function neural network and offline reinforcement learning. Specifically, they employ an extended Radial Basis Function Neural Network (RBFNN) in offline reinforcement learning algorithms as a policy network. This allows for the end-to-end learning of all parameters and network weights of RBF functions through gradient descent without needing external optimization. The proposed method was tested on five common intrusion detection datasets (UNSW-NB15, NSL-KDD, AWID, CIC IDS2017, and CIC DDOS2019) and achieved good performance indicators. However, this method may become slow when dealing with large-scale data. Additionally, offline reinforcement learning methods require storing historical data for training, which may require a large amount of storage space. Furthermore, since historical data may not fully cover all cases, the trained model may not adapt to all situations.

Recently, Xu et al. [26] introduced the Multilayer Intrusion Detection Model that employs Stacked Bidirectional Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Convolutional Denoising Autoencoders (SCDAE), and self-attention mechanisms to improve the accuracy of the model. The model also utilizes attention mechanisms to extract features and classify network traffic data. However, this model's practicality and scalability in real-world scenarios may be limited due to its requirement for a significant amount of training data and computing resources to achieve high accuracy. In another study [27], this paper explores using Graph Neural Networks (GNN) in network security intrusion detection. It focuses explicitly on applying graph representation learning techniques in intrusion detection. GNN can learn effective representations from

graph-structured data without external domain knowledge, which is crucial for effective detection. However, GNN's computational cost may be high, and it may perform poorly on small or sparse graphs because they require a certain degree of connectivity to learn effective representations. Further improvement is still needed.

Yang and Wang [28] proposed an improved wireless network intrusion detection method based on a convolutional neural network. Compared with traditional models, this method has a lower false positive rate, higher true positive rate, and higher detection accuracy. However, this method is based on specific datasets and has not been further tested and validated, which may not be suitable for real-world scenarios or other datasets. Kim and Pak [29] proposed a method for early detection of network intrusion using a type of classifier based on Generative Adversarial Networks (GAN). This method takes the packet data as the input feature and judges whether the data packet is malicious traffic through the model, enabling real-time and accurate intrusion detection without the delay time of session termination or collecting a certain number of packets. Nonetheless, the problem of dilution of intrusion features caused by timeouts and traffic outside the intrusion cycle still exists. Despite the emergence of numerous new deep learning models, most still operate by processing data on request. However, this approach overlooks the potential correlation between data requests in adjacent time frames.

In 2017, the Google team proposed the Transformer [30], which quickly became popular and was widely applied in natural language processing. The subsequent BERT [31] model has been commonly used in machine translation, text generation, and other tasks. Previous neural network models often have their limitations, such as convolutional neural network (CNN) and recurrent neural network (RNN), which struggled with processing sequential data due to issues like RNNs' gradient vanishing problem and CNNs' weak local perception ability, the Transformer overcomes these shortcomings. It supports parallel computing and better global information representation and has long-sequence solid modeling capabilities. In 2020, the Vision Transformer (ViT) [12] emerged, applying the Transformer to the field of vision and achieving excellent performance in computer vision tasks.

However, in the traditional attention mechanism, each feature calculates the relationship with the entire feature map, so using the dense self-attention mechanism in ViT will cause excessive memory and computational costs, and the features may be affected beyond the area of interest. Due to the impact of irrelevant parts, too many keys per query patch will incur high computational costs and slow convergence, increasing the risk of overfitting. Therefore, a deformable attention mechanism [16] method is proposed. Each feature only calculates the relationship with surrounding features, which reduces the computing power and avoids the influence of irrelevant feature areas.

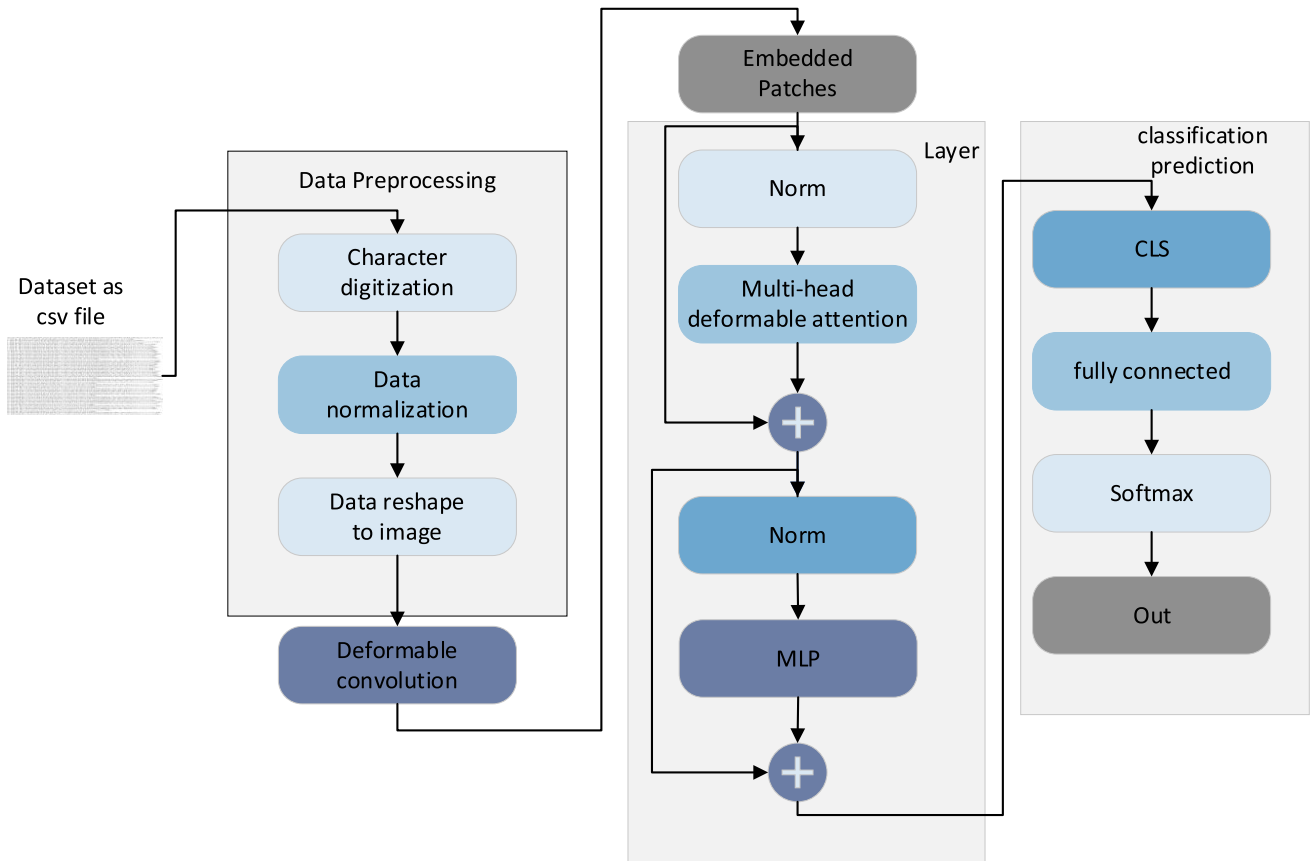


FIGURE 1. Architecture diagram of intrusion detection model.

At the same time, compared with other traditional neural networks, it also has the advantage of focusing on key information and processing long sequences. For example, compared with convolutional neural networks, the deformable attention mechanism is better able to capture global information because it can pay attention to all positions in the input sequence simultaneously, thereby better understanding the relationship between contexts, and can directly process the entire sequence without losing position information, so it is more suitable for handling dependencies between different positions. Compared with recurrent neural networks, the deformable attention mechanism allows the output of all positions to be calculated simultaneously to achieve better parallel computing and improve the training and inference efficiency of the model. Compared with graph neural networks, in addition to global information capture and parallel computing, the deformable attention mechanism can process graph-structured data and handle other types of inputs, such as sequence and set data. Therefore, it has broader application capabilities in various tasks and fields without being limited by data type. In summary, the application of the deformable attention mechanism in network intrusion detection is very suitable, and the current application of this method in network intrusion detection is scarce.

It provides a reference for other colleagues to conduct further research.

III. METHODS

Fig. 1 presents the flow chart of the complete model framework. The model has been enhanced based on the original ViT, forming a new DE-ViT model. The overall steps of the model are as follows:

Step 1: In the original datasets, the CIC IDS2017 and UNSW-NB15 datasets each consist of 80 and 48 features, respectively, represented in numerical and textual formats. Non-numeric attributes are subjected to one-hot encoding to facilitate the model's handling of these features. Additionally, rows containing Not a Number (NaN) and infinity (INF) values and irrelevant numerical columns are excluded. Following this preprocessing step, we obtain a row of 64 feature sequences of numerical values. This sequence is then reshaped into an 8×8 image matrix.

Step 2: After obtaining the feature images, undergo processing through a deformable convolution layer with a small stride of 1 and 3×3 convolution kernel to extract image features. The extracted features are then passed through an embedding layer for encoding, incorporating position encoding. Finally, the encoded features are fed into a

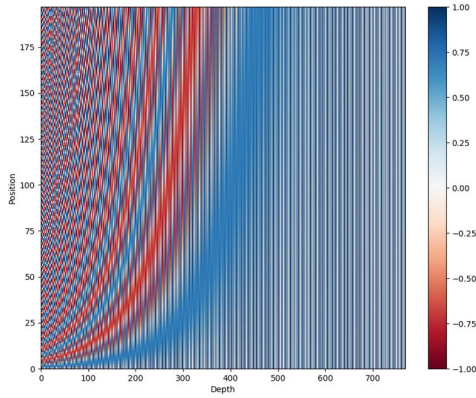


FIGURE 2. Visualization of sin-cosine positional encoding.

deformable attention layer, enabling the model to focus on specific regions of interest, thereby improving the model's comprehension of image information. After training, the model parameters retain memorized image information corresponding to each category, facilitating subsequent data classification.

A. FEATURE EXTRACTION

The earliest transformers were used to process one-dimensional sequence data. After the appearance of the Vision Transformer (ViT) [12], the application of the transformer in the image field became a reality. To achieve this, the image $X \in \mathbb{R}^{H \times W \times C}$ is partitioned into blocks to obtain $X_P \in \mathbb{R}^{N \times (P \times P \times C)}$. However, this approach may cause loss of structural information at the segmentation boundary, which is not conducive to modeling features such as image edges. A sliding window mechanism is proposed to segment the data to address this issue. Moreover, deformable convolution is used instead of regular 2D convolution to increase the receptive field of each patch. This has three benefits:

1. The proposed method utilizes adjacent position and edge information, effectively enhancing the correlation between each patch and establishing a connection with its neighbors. This approach enables the model to understand global information better and improve overall performance.
2. The ViT model's input size constraints can be relaxed using a sliding window approach to segment the image instead of strictly dividing it into equal-sized blocks. This method eliminates the requirement of dividing the image into $N = H \times W / (P \times P)$ blocks of equal size, as specified by the standard ViT model.
3. Deformable convolution [17] allows the convolution kernel to have an offset, allowing it to have a larger receptive field and better perceive global information without increasing the kernel's size.

In traditional natural language processing tasks, word position information is included, whereas in Transformer-based models, this information is discarded. However, patch position order contains important structural information for

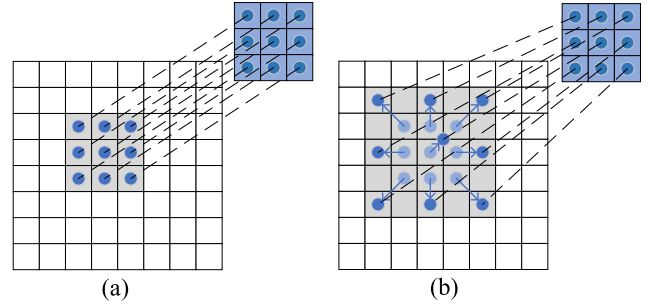


FIGURE 3. Ordinary convolution and deformable convolution.

images that must be preserved. Therefore, a sinusoidal positional encoding is proposed that embeds position information into a texture that periodically varies with the dimension index (as shown in Fig. 2). Since different positions represent different features, adding positional encoding [30] is essential for accurate feature modeling.

$$\begin{aligned} PE_{(pos, 2i)} &= \sin(pos/10000^{\frac{2i}{d_{model}}}) \\ PE_{(pos, 2i+1)} &= \cos(pos/10000^{\frac{2i}{d_{model}}}) \end{aligned} \quad (1)$$

pos is the location of patch, d_{model} is the embedding dimension of each patch, i is the location of embedding dimension.

B. DEFORMABLE CONVOLUTION

There are limitations to modeling large, unknown shape transformations using traditional CNNs. Since different locations may correspond to objects of various scales or sizes, the convolutional unit's fixed sampling of the input feature map can result in all activation units having the same receptive field. This can potentially hinder the precise localization of objects. To address this issue, deformable convolution is introduced to improve deformation modeling ability. This method is based on a parallel network that learns offset, causing the convolutional kernel to shift at the sampling points of the input feature map, thereby focusing on the region or target of interest. Fig. 3 illustrates this process.

Specifically, the traditional Transformer model utilizes 2D convolution for feature extraction, employing large convolutional kernels and strides. This approach is necessary because, for image data, even smaller kernels and strides would still yield feature maps with sizes reaching tens of thousands of pixels. Calculating attention mechanisms for each pixel would thus lead to an extremely high computational complexity. However, the features are considerably smaller for network intrusion detection than for image data. Therefore, employing small strides and convolutional kernels with continuous sliding windows can adequately extract edge information, resulting in more reliable model outcomes.

In traditional convolution, sampling is done on the regular grid R of the input feature map y , and the sum of weighted sampling values by w . R represents the relative coordinates of

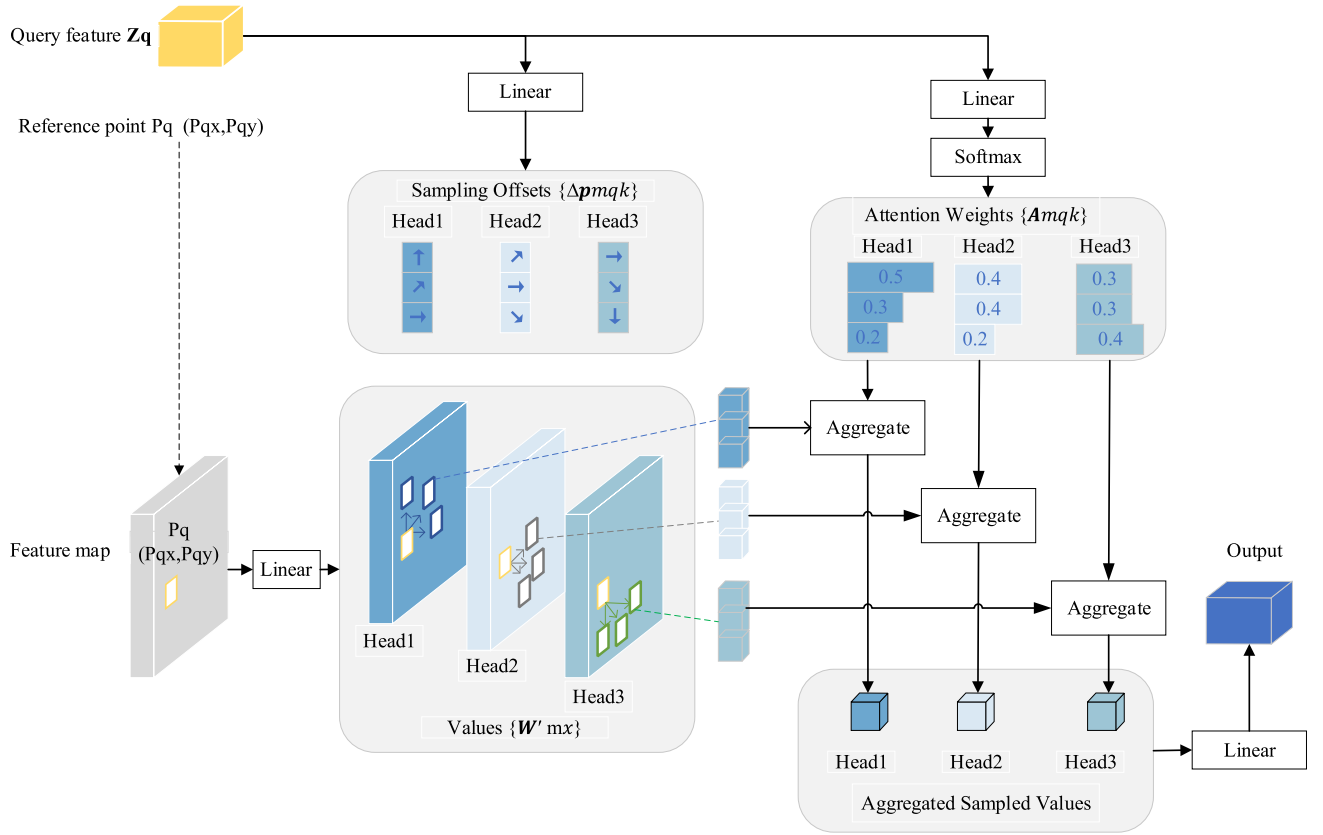


FIGURE 4. Deformable attention module.

each point in the output feature map, p_0 is a point on feature map y corresponding to the center point of the convolution kernel, and p_n is each offset of p_0 within the convolutional kernel range.

$$R = \{(-1, -1), (-1, 0), \dots, (0, 1), (1, 1)\} \quad (2)$$

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n) \quad (3)$$

Deformable convolution introduces an offset Δp_n for each point based on traditional convolution, which is generated by another convolution with the input feature map and is usually a decimal.

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \quad (4)$$

Since the offset position after addition is not an integer and does not correspond to the actual pixel points on the feature map, interpolation is needed to obtain the offset pixel value, usually using bilinear interpolation, where $p = p_0 + p_n + \Delta p_n$, and q enumerates all integral spatial positions in the feature map x . $G(\cdot, \cdot)$ represents the bilinear interpolation kernel.

$$x(p) = \sum_q G(q, p) \cdot x(q) \quad (5)$$

C. SELF ATTENTION

Let's examine the attention mechanism [30] of the Vision Transformer. The input is a flattened feature map $X \in \mathbb{R}^{N \times C}$, and it is processed by a multi-head self-attention (MHSA) block with M heads. The MHSA block can be represented as follows:

$$q = xW_q, k = xW_k, v = xW_v \quad (6)$$

$$z^{(m)} = \sigma(q^{(m)}k^{(m)T} / \sqrt{d})v^{(m)} \quad m = 1, \dots, M \quad (7)$$

$$z = \text{Concat}(z^{(1)}, z^{(2)}, \dots, z^{(M)})W_o \quad (8)$$

Here, $\sigma(\cdot)$ denotes the softmax function, and $d = C/M$ represents the dimension for each head. $z^{(m)}$ represents the embedding output of the m -th attention head, while $q^{(m)}, k^{(m)}, v^{(m)} \in \mathbb{R}^{N \times d}$ represent the query embedding, key embedding, and value embedding respectively. The projection matrices $W_q, W_k, W_v, W_o \in \mathbb{R}^{C \times C}$ are used in the MHSA block. To introduce non-linearity, a Transformer block typically includes an MLP block with two linear transformations and a GELU activation.

By incorporating normalization layers and identity shortcut connections, the l -th transformer block can be expressed as follows:

$$z'_l = \text{MHSA}(\text{LN}(z_{l-1})) + z_{l-1} \quad (9)$$

$$z_l = \text{MLP}(\text{LN}(z'_l)) + z'_l \quad (10)$$

D. DEFORMABLE ATTENTION

In traditional self-attention mechanisms, obtaining the feature vector at each position involves the weighted summing of the entire feature map. However, in deformable attention mechanisms, the feature vector at each position is obtained by weighted summing of the feature vectors of adjacent small blocks. That can be translated, rotated, scaled, and transformed in different directions. This deformable approach enables the model to better adapt to objects of different shapes and sizes and to capture interactions between objects more accurately.

Why do traditional attention mechanisms consume more memory and computational costs? One major challenge with using Transformer attention for image feature mapping is that it considers all possible spatial positions, leading to computational inefficiency. To overcome this limitation, a deformable attention mechanism has been proposed as an alternative to the traditional attention mechanism. This approach draws inspiration from deformable convolution and focuses on a small number of key sampling points around the reference point, as illustrated in Fig. 4. By assigning a small fixed number of keys to each query, not only can the computational resources of the model be reduced, but also the problems related to convergence and feature space resolution can be alleviated. To enhance comprehension, Fig. 4 displays the input data as an image. However, the input image is flattened into a sequence input.

Specifically, the first step is to determine the number of your sampling points, perform two full connections on the feature map, and determine the offsets and weights relative to the sampling points around the feature points. The weights and offsets are random at the beginning; during the model training, the obtained weights and offsets are more and more consistent with the optimization results, focusing on the area of interest and ignoring non-correlated areas so that each feature only calculates the relationship between a small number of surrounding feature points, which not only reduces It reduces computing power and memory, and avoids the impact of irrelevant areas on the results.

The input feature map $x \in \mathbb{R}^{H \times W \times C}$ is given. we use q to index a query element with content feature z_q and a two-dimensional reference point P_q . The deformable attention mechanism can be calculated using the following formula:

$$\begin{aligned} \text{DeformAttn}(z_p, p_q, x) \\ = \sum_{m=1}^M W_m \left[\sum_{k=1}^K A_{mqk} \cdot W'_m x(p_q + \Delta p_{mqk}) \right] \end{aligned} \quad (11)$$

In the equation, we denote the total number of attention heads as M and the total number of sampled keys as K , where K is always less than HW . Additionally, we use Δp_{mqk} and A_{mqk} to denote the sampling offsets and attention weights of the m -th attention head and k -th sampled point, respectively. The attention weight A_{mqk} ranges from 0 to 1, $\sum_{k=1}^K A_{mqk} = 1$. $\Delta p_{mqk} \in \mathbb{R}$ is a two-dimensional unconstrained real number. As $P_q + \Delta p_{mqk}$ is a fractional order, bilinear interpolation

is utilized to compute the precise value following the offset. Δp_{mqk} and A_{mqk} are both derived from linear projection on the query feature Z_q , with W_m and W'_m serving as projection matrices. Specifically, the query feature Z_q is inputted into a linear projection operator with $3MK$ channels. The first $2MK$ channels encode the sampling offset Δp_{mqk} , while the remaining MK channels are directed to the Softmax operator to obtain the attention weight A_{mqk} .

This is because every sampling point in each attention head possesses K sampling keys. Each sampling key necessitates the prediction of horizontal and vertical coordinates for the offset, while the weight solely requires the prediction of a singular value.

The procedure entails the addition of offset A_{mqk} to the two-dimensional reference point P_q to acquire the offset coordinates. The value of this offset is subsequently found in feature map x , based on these coordinates. The offset coordinates are then multiplied by attention weight A_{mqk} and ultimately output through a fully connected layer.

E. LOSS FUNCTION

To address the issue of imbalanced intrusion detection datasets, we propose a layered focal loss function, called L-Focal loss. Firstly, we start with the binary classification cross-entropy loss and gradually introduce the L-Focal loss function.

where y is the label of the sample, and $y' \in [0, 1]$ is the estimated probability of the model for the class with label $y = 1$. p is the final probability

$$p = \begin{cases} y', & \text{if } y = 1, \\ 1 - y', & \text{otherwise.} \end{cases} \quad (12)$$

$$CE(p, y) = CE(p) = -\log(p) \quad (13)$$

In intrusion detection datasets, there is an imbalance in the different categories of data, with normal traffic samples being much more abundant than other types of attack samples. Negative samples account for the dominant part of the total loss. The model predicts attack traffic as normal traffic to reduce loss and improve accuracy. This makes model optimization unsatisfactory. This violates our original intention, but the L-Focal loss function assigns the weight of positive and negative samples in the total loss by varying the weight factor α . The weight decreases as the number of samples increases, resulting in a more minor impact on the loss function.

$$BCE(p, y, \alpha) = \begin{cases} -\alpha \log(p), & \text{if } y = 1, \\ -(1 - \alpha) \log(p), & \text{otherwise.} \end{cases} \quad (14)$$

When too many negative samples are easily distinguished, the entire training revolves around these negative samples, overwhelming positive samples and causing significant loss. Therefore, we introduce a modulation factor γ to focus on complex samples. In the L-Focal loss function, γ is utilized as a modulation factor to amplify the loss of challenging-to-classify samples and reduce the impact of easily classifiable

samples on the model. However, entering different modulation factor parameters results in significant differences in output results, leading to overfitting problems when there are outliers.

It is not uncommon for an intrusion detection model to find rare types of attacks in a dataset [23]. However, directly using $(1-p)^\gamma$ as a modulation factor for the loss function during the model training process can lead to continuous loss reduction without improving the model's accuracy. This may not be conducive to model development and research. The L-Focal loss function, on the other hand, classifies samples into different levels of classification, with a focus on accurately classifying samples near $p = 0.5$, rather than overemphasizing highly classifiable samples with $p \gg 0.5$ or outliers with $p \ll 0.5$.

$$g(p) = \begin{cases} (1-p)^{\gamma_1}, & |p-0.5| < a, \\ (1-p)^{\gamma_2}, & \text{otherwise}, \end{cases} \quad (15)$$

$$LFocal = \begin{cases} -\alpha g(p) \log p, & \text{if } y = 1, \\ -(1-\alpha) g(p) \log p, & \text{otherwise}. \end{cases} \quad (16)$$

To achieve this, the L-Focal loss function introduces a modulation factor, $g(p)$, and a weighting factor, $\alpha \in [0, 1]$, while $p \in [0, 1]$ represents the estimated probability of the model. By setting a value to classify samples in the L-Focal loss function, we can focus on samples between $[0.5-a, 0.5+a]$ during training. This approach helps prevent overfitting or large cumulative sample losses that can dominate gradients and adversely affect model optimization. To ensure that the model focuses on appropriate samples during training, we set $0 \leq a \leq 0.5$ and $0 \leq \gamma_1 < \gamma_2 \leq 5$ in the L-Focal loss function.

IV. EXPERIMENT

A. DATASET

In this study, we utilized two datasets, CIC IDS2017 [32] and UNSW-NB15 [33], due to their diverse attack types and relevant features essential for server identification and arrival time.

The CIC IDS2017 and UNSW-NB15 datasets are publicly available datasets downloaded from official websites. The CIC IDS2017 dataset is provided in CSV format and includes timestamps, source and destination IP addresses, source and destination ports, protocol, and other information.

These datasets encompass benign daily activities as well as various types of attacks, such as Benign, DoS (Denial of Service), Heartbleed, DDoS (Distributed Denial of Service), Patator, Web attack, Bot, PortScan, and Infiltration. The UNSW-NB15 dataset is provided in CSV format and includes timestamps, source and destination IP addresses, source and destination ports, protocol, and response speed, among other details. These datasets encompass benign daily activities and attacks, such as Normal, DoS (Denial of Service), Generic, Reconnaissance, Analysis, Exploits, Fuzzers, Backdoor, Shellcode, and Worms.

As stated in the beginning of part III, CIC IDS2017 has 80 features, while UNSW-NB15 has 48 features. To preprocess the data, we one-hot encoded the characters, removed columns with identical values, cleaned some dirty data, processed it into 64 numerical features, normalized it, used the matplotlib library to visualize images, reshaped it into 8×8 image data as shown in Fig. 6 and Fig. 7, and finally classified it using a model.

Table 2 presents the positive and negative sample classification of CIC IDS2017 [32] and UNSW-NB15 [33] datasets. While Fig. 5 shows the distribution of various data types in these datasets.

B. EVALUATION

There are several metrics available to evaluate NIDS performance, including False Positive (FP), False Negative (FN), True Negative (TN), and True Positive (TP). We will use the following metrics to test NIDS based on these parameters:

Precision (P): This metric calculates the percentage of attack samples correctly identified out of all samples classified as attacks.

$$P = \frac{TP}{TP + FP} \quad (17)$$

Recall Rate (RC): This metric calculates the percentage of attack samples that were correctly identified across the entire attack sample

$$RC = \frac{TP}{TP + FN} \quad (18)$$

Accuracy (ACC): This represents the percentage of correctly classified samples among all normal and abnormal error instances.

$$ACC = \frac{TN + TP}{TN + TP + FN + FP} \quad (19)$$

False positive Rate (FPR): This measures the proportion of normal samples judged as attacks among all normal samples.

$$FPR = \frac{FP}{FP + TN} \quad (20)$$

C. EXPERIMENTAL RESULTS

The DE-ViT model has an input image size of 8×8 , with a patch size of 3×3 , 12 heads for multi-head self-attention, and a hidden layer size of 768. It consists of 12 layers and an MLP size of 3072. The deformable attention mechanism samples 8 points surrounding each reference point, and the deformable convolutional kernel size is 3×3 with a stride of 1; therefore, the number of patches is 36. The learning rate uses a cosine annealing decay strategy.

In the L-Focal loss function, $\alpha = 0.65$, $\gamma_1 = 2$, $\gamma_2 = 3$, and $a = 0.3$. The model is compared with a regular ViT model in experiments. The structural improvements are shown in Table 1.

During the study, the data was classified as benign and malicious. The dataset was divided into 20% for testing, 10% for validation, and 70% for training. Fig. 9 shows the training

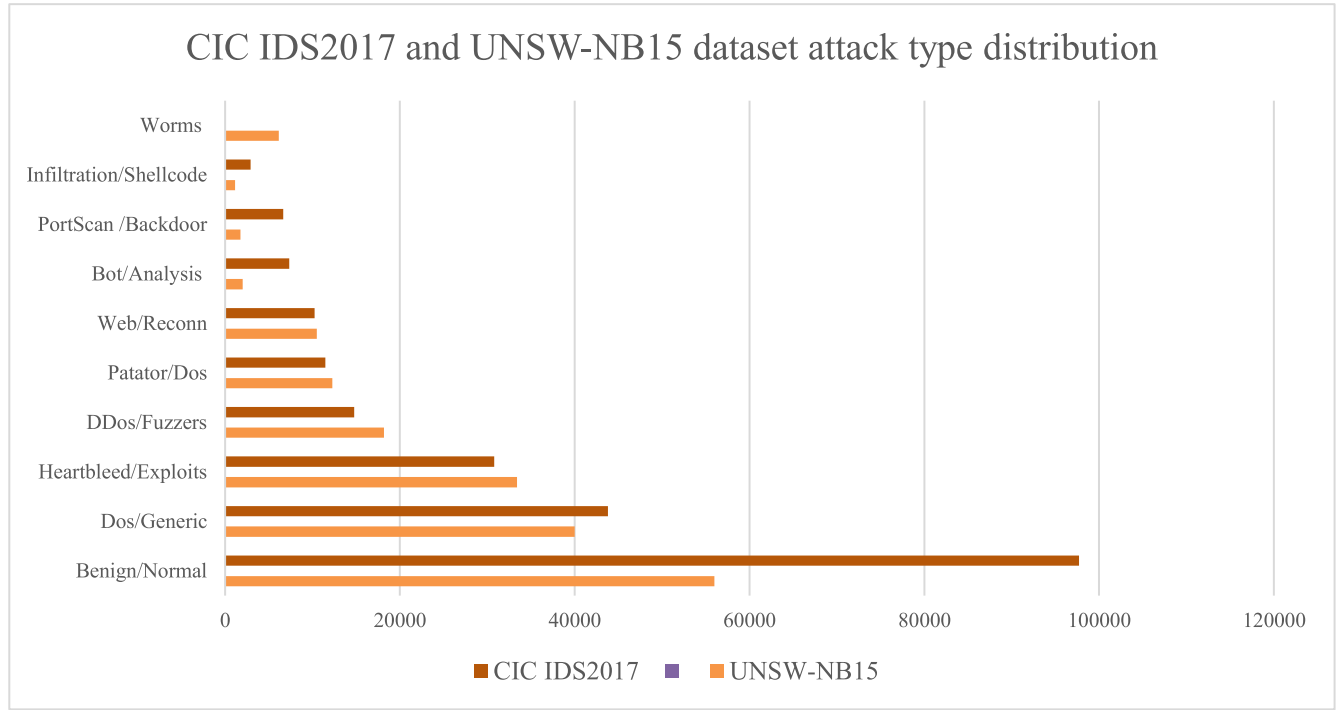


FIGURE 5. The number of attack types comes from CIC IDS2017 and UNSW-NB15.

TABLE 1. VIT and DE-VIT model comparison.

Model	Sine-Cosine Position Encoding	cosine annealing	L-Focal loss	DeformConv2d	Deformable Attention
VIT	×	×	×	×	×
DE-VIT	√	√	√	√	√

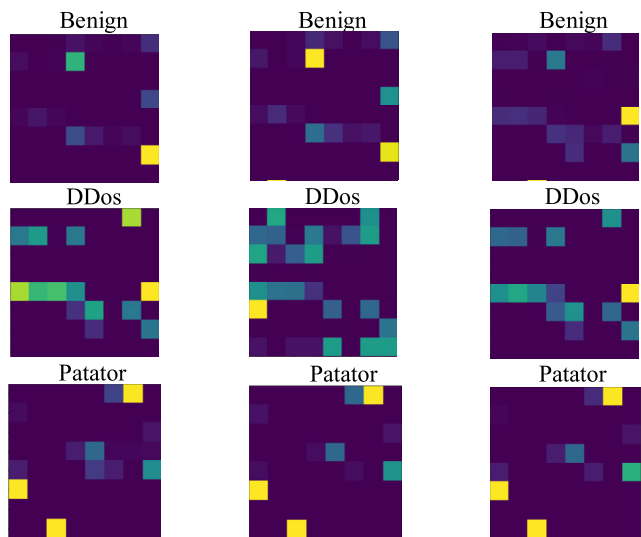


FIGURE 6. CIC IDS2017 dataset attack type data visualization.

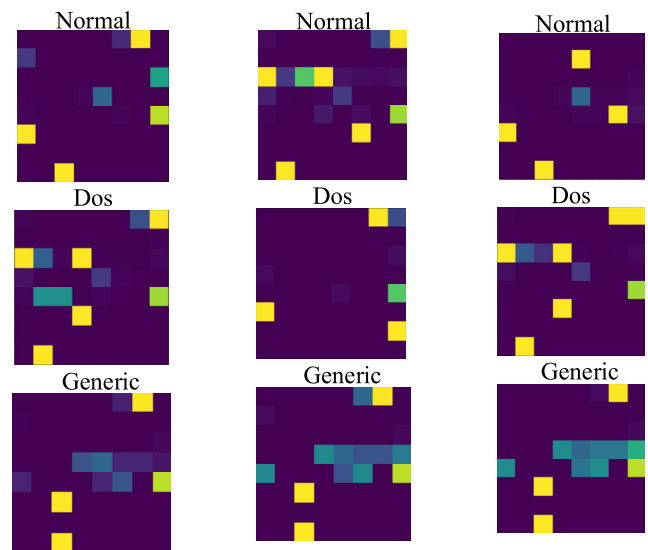


FIGURE 7. UNSW-NB15 dataset attack type data visualization.

curve of the DE-VIT model on the CIC IDS2017 dataset, while Table 3 presents the False Positive Rate (FPR) and

Accuracy (ACC) results for both the CIC IDS2017 [32] and UNSW-NB15 [33] datasets.

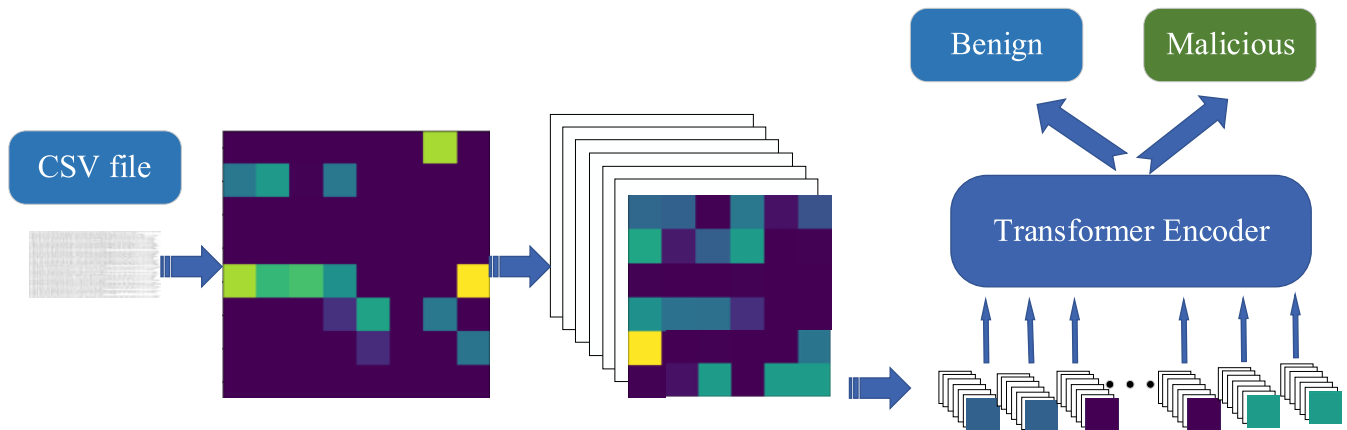


FIGURE 8. The overall structure of our system.

TABLE 2. The number of positive and negative samples comes from CIC IDS2017 and UNSW-NB15 datasets.

Dataset	Benign	Malicious
UNSW-NB15	56000	119341
CIC IDS2017	97718	127993

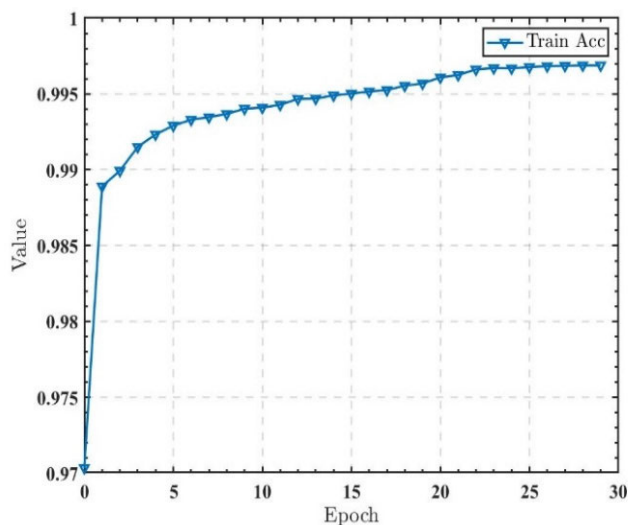


FIGURE 9. DE-VIT training curve on CIC IDS2017 dataset.

TABLE 3. The final accuracy, false positive rates and recall rates.

	VIT			DE-VIT		
Dataset	ACC	FPR	Recall	ACC	FPR	Recall
UNSW-NB15	95%	4.2%	93.6%	97.2%	1.4%	94.89%
CIC IDS2017	97.4%	1.9%	96.6%	99.5%	0.3%	99.3%

Based on the experimental results, it has been demonstrated that DE-VIT outperforms VIT on various datasets

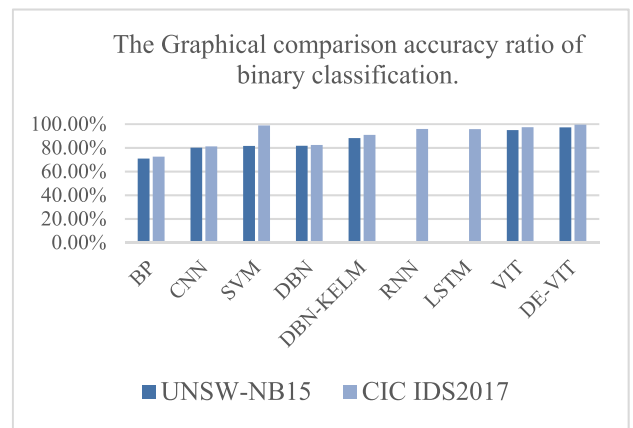


FIGURE 10. The Graphical comparison accuracy ratio of binary classification.

TABLE 4. Comparison with other methods for binary classification.

DATASET	UNSW-NB15		CIC IDS2017	
	Precision	Accuracy	Precision	Accuracy
BP [34]	59.90%	70.89%	63.24%	72.56%
CNN [35]	81.29%	80.11%	83.49%	81.22%
SVM [25]	75.2%	81.5%	74.5%	98.9%
DBN [34]	73.13%	81.69%	75.68%	82.39%
DBN-KELM [34]	80.61%	88.21%	83.69%	90.95%
RNN [24]			95.86%	95.83%
LSTM [24]			95.82%	95.79%
VIT	92.1%	95.0%	97.5%	97.4%
DE-VIT	97.1%	97.25%	99.59%	99.5%

regarding recall rate, accuracy, and precision. Therefore, DE-VIT is considered a superior model compared to VIT. In addition, the results suggest that DE-VIT can provide better performance in various computer vision tasks. These findings are significant for researchers and practitioners in computer vision, as they can utilize DE-VIT to improve their models' accuracy and enhance their applications' performance.

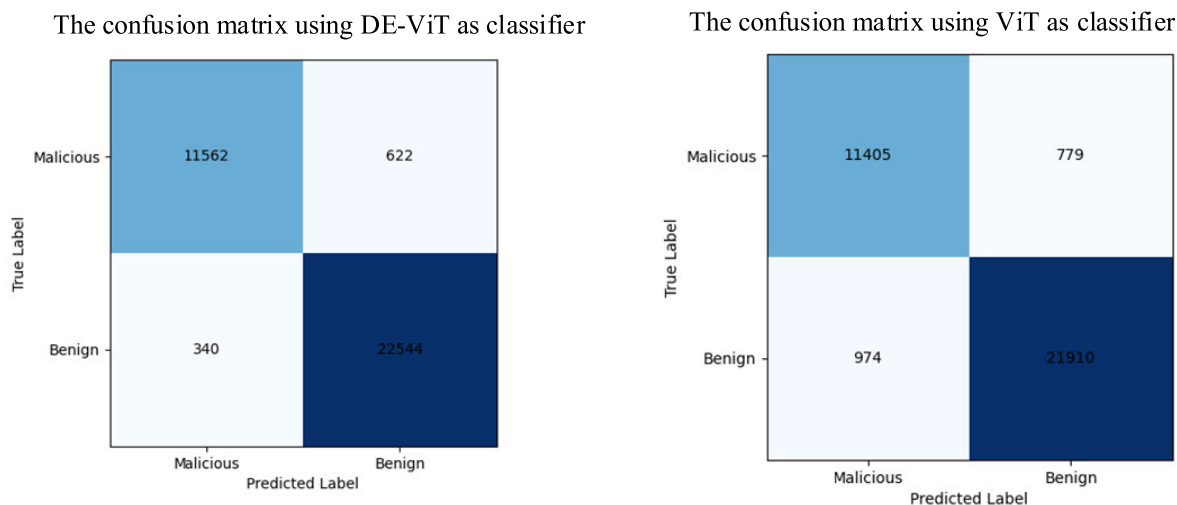


FIGURE 11. The confusion matrices using the ViT and DE-ViT architectures for binary classification on the UNSW-NB15 dataset.

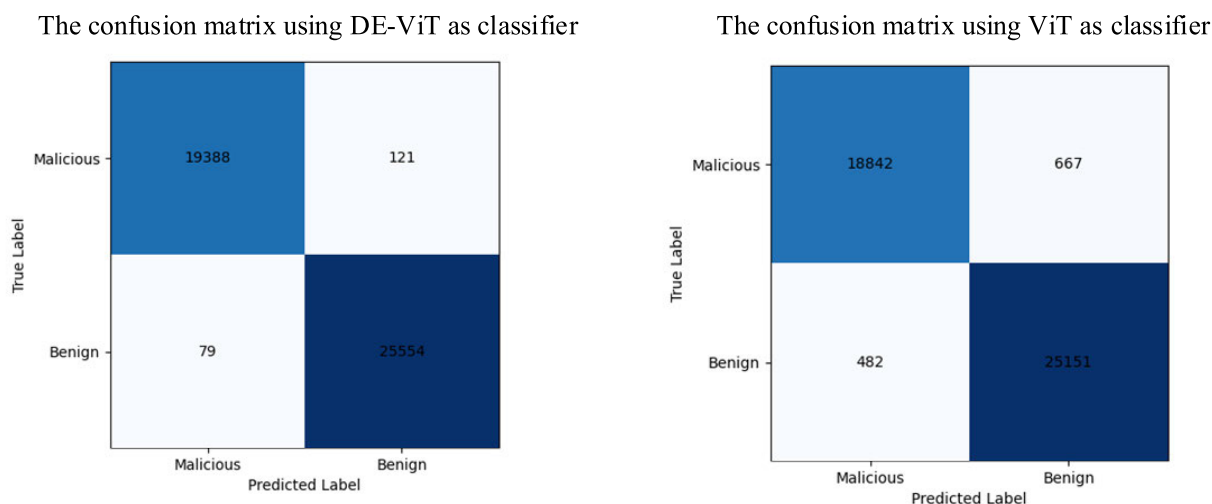


FIGURE 12. The confusion matrices using the ViT and DE-ViT architectures for binary classification on the CICIDS2017 dataset.

D. BINARY CLASSIFICATION EXPERIMENT

Table 4 provides a comparison of our proposed method with other NIDS methods that use machine learning or deep learning for binary classification tasks, such as BP [34], CNN [35], SVM [25], DBN [34], Deep Belief Network with Improved Kernel-Based Extreme Learning (DBN-KELM) [34], RNN [24], and LSTM [24]. The confusion matrix for binary classification on the UNSW-NB15 [32] and CIC IDS2017 [33] datasets for both ViT and DE-ViT architectures is shown in Fig. 11 and Fig. 12, respectively. With the utilization of 20% of the data for testing.

Furthermore, the accuracy comparison graph for binary classification is presented in Fig. 10. We found that our method utilizing the DE-ViT classifier outperforms all other methods in terms of accuracy and precision on both

the UNSW-NB15 [33] and CIC IDS2017 [32] datasets. Specifically, we achieved 99.5% accuracy on the CIC IDS2017 dataset and 97.25% on the UNSW-NB15 dataset. Additionally, our ViT model utilizing this method also showed promising results.

DE-ViT can outperform other algorithms; firstly, the superior performance of DE-ViT in binary classification primarily stems from the optimization of the model. In Part II, we conducted sufficient research on related work, discussing the shortcomings of most current deep learning algorithms and highlighting the advantages of our adopted deformable attention mechanism module. Additionally, we upgraded the model's feature extraction module by employing deformable convolutions, which better fulfills the requirements of feature extraction. Furthermore, we optimized the loss function to achieve a better balance in the dataset. Secondly,

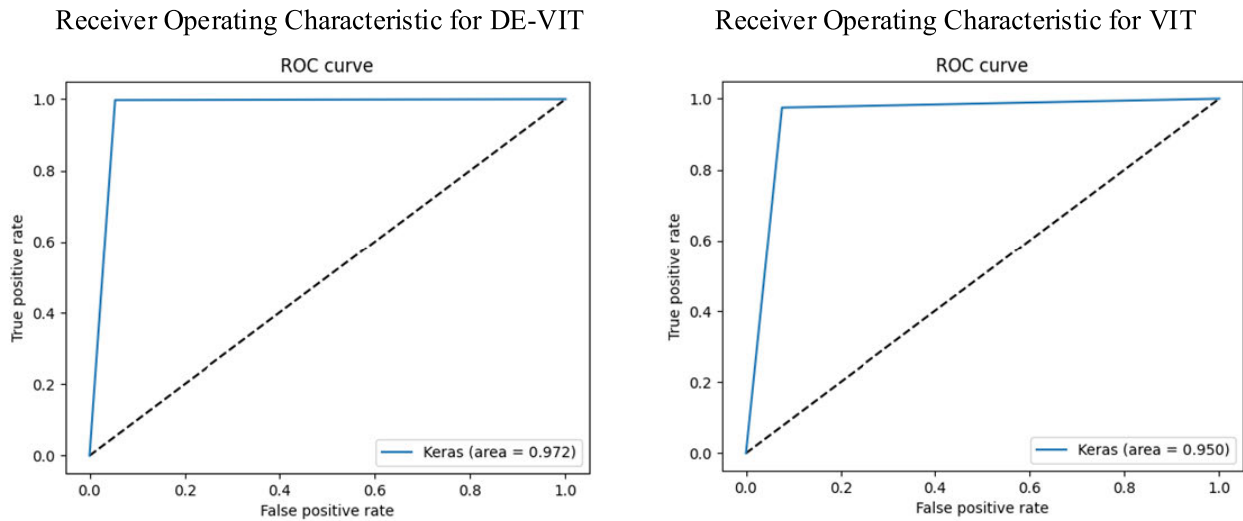


FIGURE 13. Receiver operating characteristic curve classifiers in the UNSW-NB15 dataset.

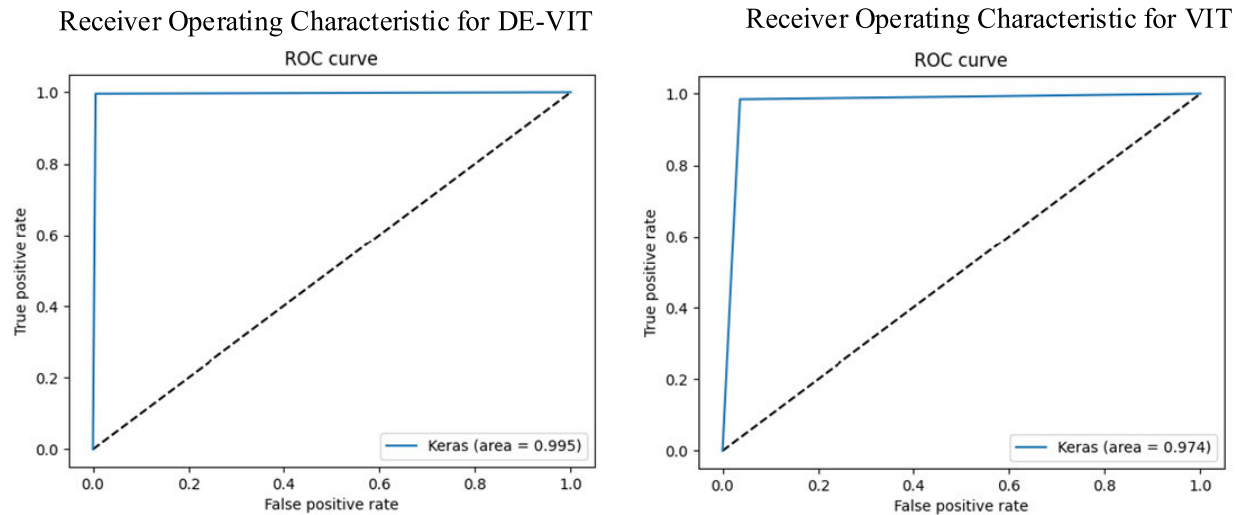


FIGURE 14. Receiver operating characteristic curve classifiers in the CIC IDS2017 dataset.

regarding the achieved accuracies of 99.5% (CIC IDS2017) and 97.25% (UNSW-NB15), the CIC IDS2017 dataset is relatively cleaner, and experts have established more reasonable feature extraction criteria based on traffic indicators. Therefore, overall, the accuracy is higher than that of the UNSW-NB15 dataset.

To visualize our findings, we have included the Receiver Operating Characteristic (ROC) curves for both the VIT and DE-VIT classifiers on both datasets in Fig. 13 and Fig. 14. Overall, Thanks to the optimization of the model structure by DE-VIT, satisfactory results are achieved on both datasets. Our research demonstrates the effectiveness of our approach in accurately classifying data in these two datasets.

E. MULTI-CLASSIFICATION EXPERIMENT

We conducted multi-class classification experiments on the UNSW-NB15 and CIC IDS2017 datasets [32], which have imbalanced generated image datasets. In the UNSW-NB15 dataset [33], Normal accounts for 31% of the data, while in the CIC IDS2017 dataset, benign accounts for 43%. This indicates that normal data types make up most, while attack data types are relatively rare.

Table 5 displays the performance of the DE-VIT classifier on the multi-class test set of the UNSW-NB15 dataset, which has ten categories: Normal, Dos, Generic, Reconnaissance, Analysis, Exploits, Fuzzers, Backdoor, Shellcode, and Worms. The test set constitutes 20% of the entire dataset. Following one-hot encoding of the categories, a multi-class

TABLE 5. Performance of the DE-VIT classifier on the test set for multi-class classification on the UNSW-NB15 dataset.

	Normal	Generic	Exploits	Fuzzers	Dos
Numbers	12936	7729	5132	2062	3089
Recall	96.6%	98.9%	94.9%	88.3%	89.4%
Precision	98.3%	96.9%	96.9%	90.2%	92.5%
Accuracy	97.1%	95.1%	95.1%	92.4%	93.6%
	Reconnaissance	Analysis	Backdoor	Shellcode	Worms
Numbers	2496	577	483	378	44
Recall	97.1%	84.6%	85.4%	79.6%	70.2%
Precision	98.1%	82.5%	90.1%	71.1%	78.3%
Accuracy	97.8%	89.7%	92.3%	80.6%	72.7%

TABLE 6. Comparison with other methods for multi-class classification.

	Precision					Acc
	Normal	Generic	Exploits	Fuzzers	Dos	
CNN [35]	82.1%	75.8%	77.3%	81.6%	74.6%	
DBN [34]	78.9%	90.2%	85.6%	64.4%	25.8%	
DBN-KELM [34]	89.0%	92.3%	86.9%	65.2%	30.9%	
VIT	93.6%	92.9%	91.1%	89.5%	90.3%	
DE-VIT	98.3%	96.9%	96.9%	90.2%	92.5%	
	Reconnaissance	Analysis	Backdoor	Shellcode	Worms	
CNN [35]	71.3%	75.2%	68.5%	65.5%	66.3%	80.51%
DBN [34]	66.9%	25.8%	0.5%	7.9%	2.2%	68.6%
DBN-KELM [34]	67.6%	36.8%	0.0%	11.3%	2.5%	75.6%
VIT	85.3%	77.4%	69.6%	73.2%	70.6%	90.9%
DE-VIT	98.1%	82.5%	90.1%	71.1%	78.3%	95.4%

regression test is performed to ascertain the number of accurately classified instances. Subsequently, evaluation metrics are aggregated for comprehensive statistical analysis. Each data category’s recall rate, precision, and accuracy are listed. As Normal samples have the largest quantity, the accuracy is highest, whereas categories with fewer samples have slightly lower accuracy. Table 6 compares the overall accuracy and category precision of DE-VIT with other multi-class algorithms on the UNSW-NB15 dataset. DE-VIT has an accuracy 15% higher than the second-ranked CNN and significantly outperforms other algorithms.

Table 7 shows the performance of the DE-VIT classifier on the multi-class test set of the CIC IDS2017 dataset,

which has nine categories: Benign, Dos, Heartbleed, DDos, Patator, Web attack, Bot, PortScan, and Infiltration. The recall rate, precision, and accuracy of each data category are listed, and it can be seen that each category has a higher accuracy. Table 8 compares the overall accuracy and category precision of DE-VIT with other multi-class algorithms on the CIC IDS2017 dataset [32]. The overall accuracy is 6.5% higher than the second-ranked DBN-KELM [34] and superior to other algorithms. However, precision and accuracy decrease rapidly in categories with relatively few samples. Although we have taken measures to address the imbalance issue in the data and achieved fairly good results, it is not perfect and still needs further improvement.

TABLE 7. Performance of the DE-VIT classifier on the test set for multi-class classification on the CIC IDS2017 dataset.

	Benign	Dos	Heartbleed	DDos	Patator
Numbers	15525	9524	7238	3424	2941
Recall	97.5%	97.9%	98.1%	96.3%	98.6%
Precision	99.0%	98.3%	97.5%	95.0%	98.9%
Accuracy	98.8%	98.2%	98.6%	97.6%	98.8%
	Web attack	Bot	PortScan	Infiltration	
Numbers	2344	2045	1549	552	
Recall	92.1%	89.4%	79.8%	75.6%	
Precision	93.5%	92.2%	81.3%	77.5%	
Accuracy	92.4%	89.7%	87.5%	86.9%	

TABLE 8. Comparison with other methods for multi-class classification.

	Precision					Acc
	Benign	Dos	Heartbleed	DDos	Patator	
DBN [34]	75.9%	80.6%	82.3%	69.2%	75.2%	
DBN-KELM [34]	77.8%	86.4%	85.9%	62.4%	79.4%	
VIT	95.6%	93.8%	92.3%	91.7%	95.5%	
DE-VIT	99.0%	98.3%	97.5%	95.0%	98.9%	
	Web attack	Bot	PortScan	Infiltration		
DBN [34]	82.4%	64.7%	73.2%	73.4%		84.9%
DBN-KELM [34]	79.9%	73.8%	76.4%	76.3%		90.8%
VIT	90.2%	86.8	80.9%	73.6%		93.8%
DE-VIT	93.5%	92.2%	81.3%	77.5%		97.3%

V. CONCLUSION

This paper presents a cutting-edge approach, DE-VIT, to address the challenge of intrusion detection. By transforming intrusion detection data into image data, DE-VIT employs image classification algorithms to resolve the issue effectively. The method refines the original VIT and incorporates a more advanced deformable attention mechanism instead of self-attention. An enhanced sliding window mechanism is introduced to optimize the model further for superior edge information extraction. At the same time, deformable convolution is utilized to extract edge information and expand the receptive field of each patch. Sine and cosine positional encoding are integrated to represent diverse features at varying positions. This results in a texture imbued with position information as the position embedding changes gradually with the dimension sequence number.

DE-VIT introduces an innovative deformable attention mechanism module that selects the position of key-value pairs in a data-dependent manner. This agile solution enables the deformable attention mechanism module to concentrate on pertinent areas and capture more information features, circumventing excessive memory and computational costs. To tackle the issue of imbalanced datasets, a layered focal loss function, dubbed the L-Focal loss function, is proposed to center attention on challenging-to-classify samples, thereby enhancing classification accuracy.

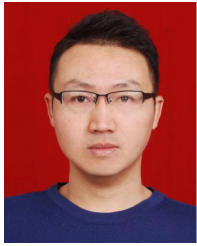
Experimental results demonstrate DE-VIT's superiority over other algorithms. In binary classification experiments, we attained 99.5% accuracy on the CIC IDS2017 dataset [32] and 97.25% on the UNSW-NB15 dataset [33], outperforming most mainstream algorithms such as CNN, LSTM, and DBN-KELM. Our algorithm surpassed others in multi-classification experiments, achieving commendable accuracy

and precision across most categories. The accuracy of small samples did not significantly decline under the constraint of the loss function, culminating in satisfactory experimental results.

Several promising research directions emerge based on the DE-ViT approach proposed in this paper. Firstly, exploring its applicability in diverse image-based classification tasks beyond intrusion detection, such as medical imaging or satellite analysis, can provide valuable insights into its versatility. Additionally, refining the deformable attention mechanism for dynamic focus adjustment and incorporating reinforcement learning techniques can enhance adaptability. Extending the application of the L-Focal loss function beyond imbalanced datasets or investigating novel loss functions can improve model robustness. Moreover, integrating DE-ViT with advanced techniques like graph neural networks or self-supervised learning methods may synergistically enhance intrusion detection performance. Exploring these directions has the potential to advance state-of-the-art intrusion detection and beyond.

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